

Multiple Choice

1. **Answer:** B
Options: A) True, True; B) False, False; C) True, False; D) False, True
Note: Correct option: B - False, False
2. **Answer:** D
Options: A) multiplication is not a binary operation; B) multiplication is not associative; C) identity element does not exist; D) zero has no inverse
Note: Correct option: D - zero has no inverse
3. **Answer:** C
Options: A) True, True; B) False, False; C) True, False; D) False, True
Note: Correct option: C - True, False
4. **Answer:** B
Options: A) 0; B) 1; C) 2; D) 3
Note: Correct option: B - 1
5. **Answer:** D
Options: A) True, True; B) False, False; C) True, False; D) False, True
Note: Correct option: D - False, True

True / False

6. **Answer:** False
Options: A) True; B) False
Note: LLM-generated false statement. Correct answer: '0'.
7. **Answer:** True
Options: A) True; B) False
Note: LLM-generated true statement based on MCQ.
8. **Answer:** False
Options: A) True; B) False
Note: LLM-generated false statement. Correct answer: 'True, True'.
9. **Answer:** True
Options: A) True; B) False
Note: LLM-generated true statement based on MCQ.
10. **Answer:** False
Options: A) True; B) False
Note: LLM-generated false statement. Correct answer: 'False, True'.

Long Form Response

11. **Answer:** f has all of its relative extrema on the line $x = y$. Explain why this is the correct answer and provide supporting evidence. Show the calculation or reasoning that leads to this conclusion.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

12. **Answer:** One. Explain why this is the correct answer and provide supporting evidence. Show the calculation or reasoning that leads to this conclusion.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.